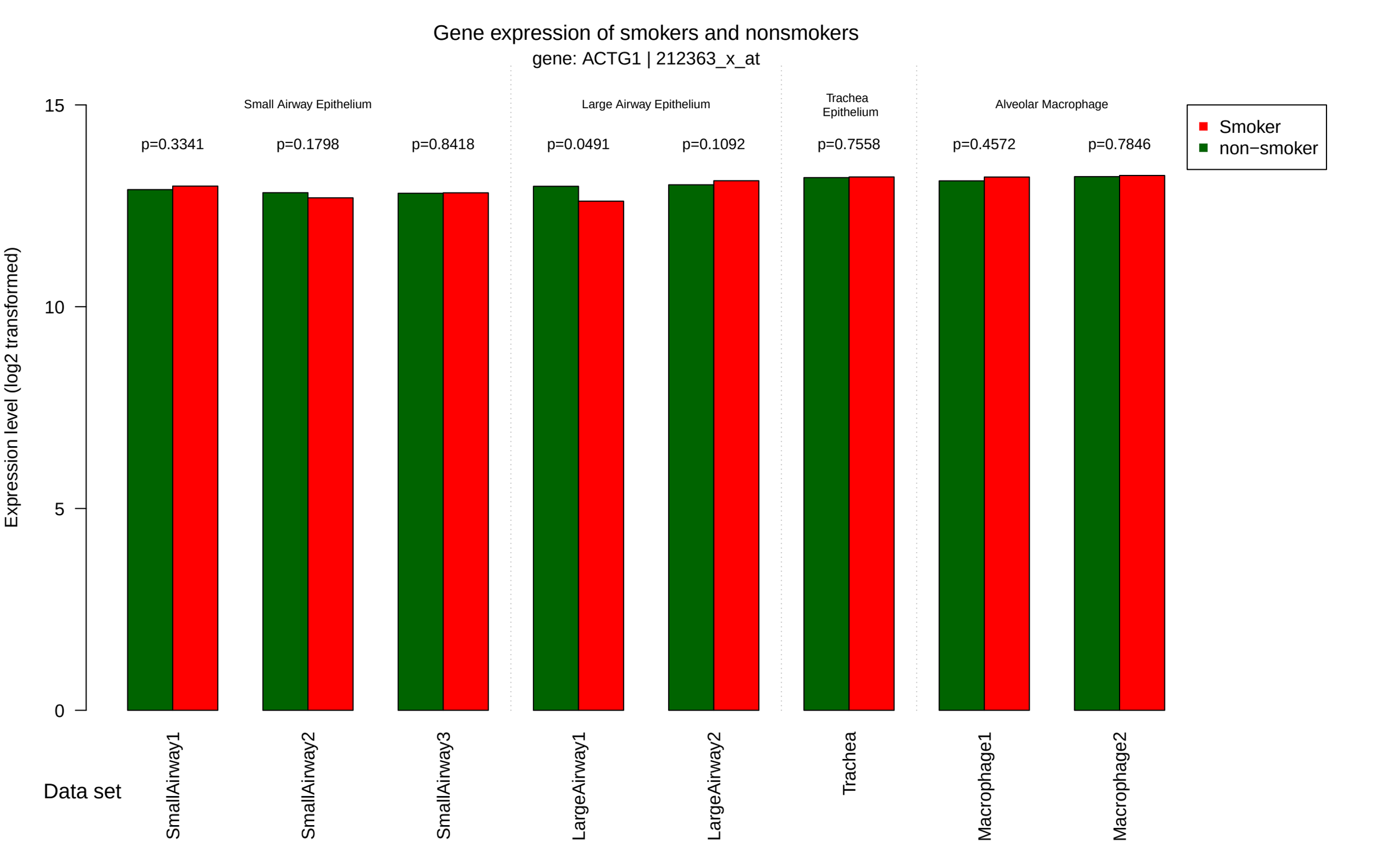
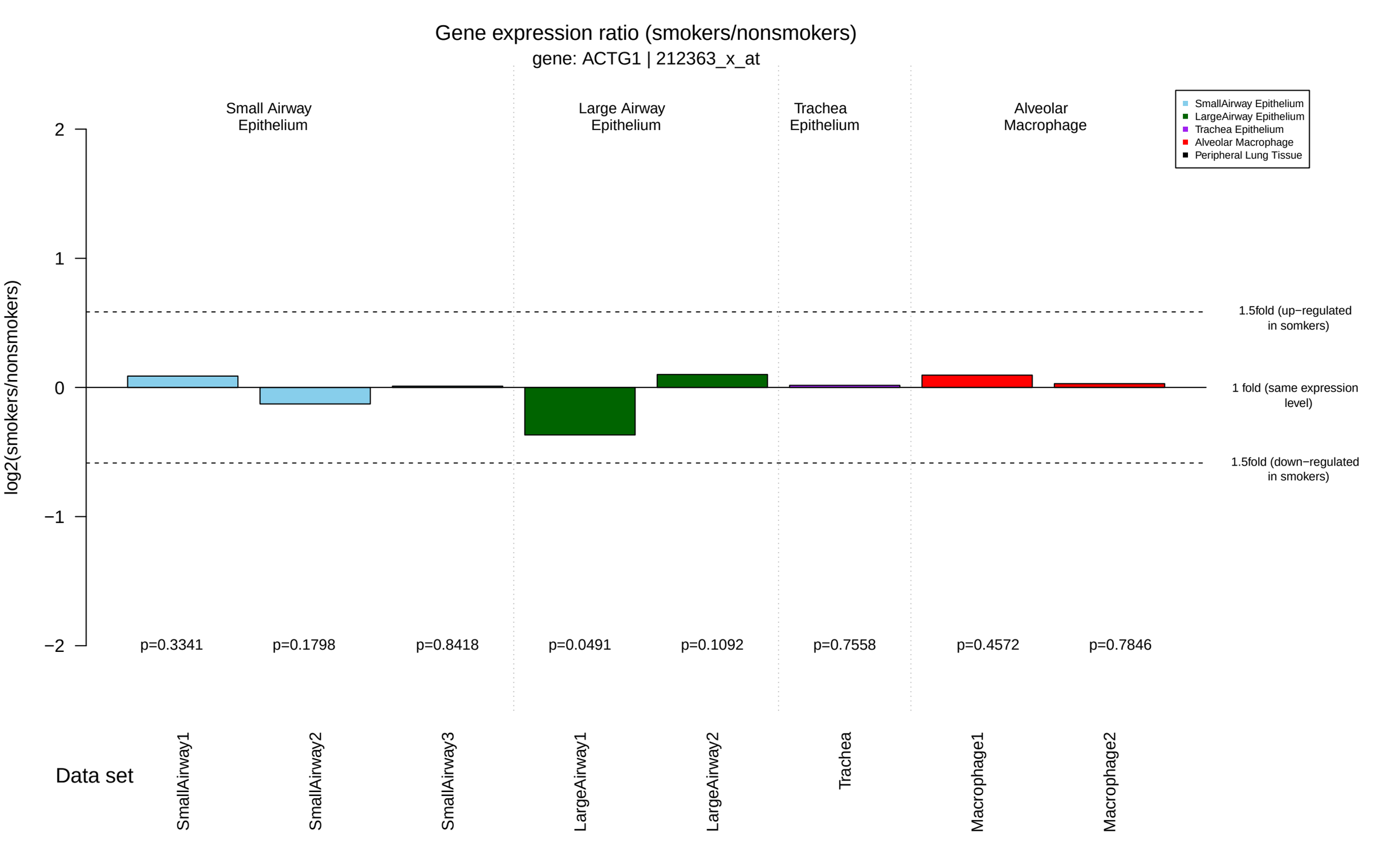
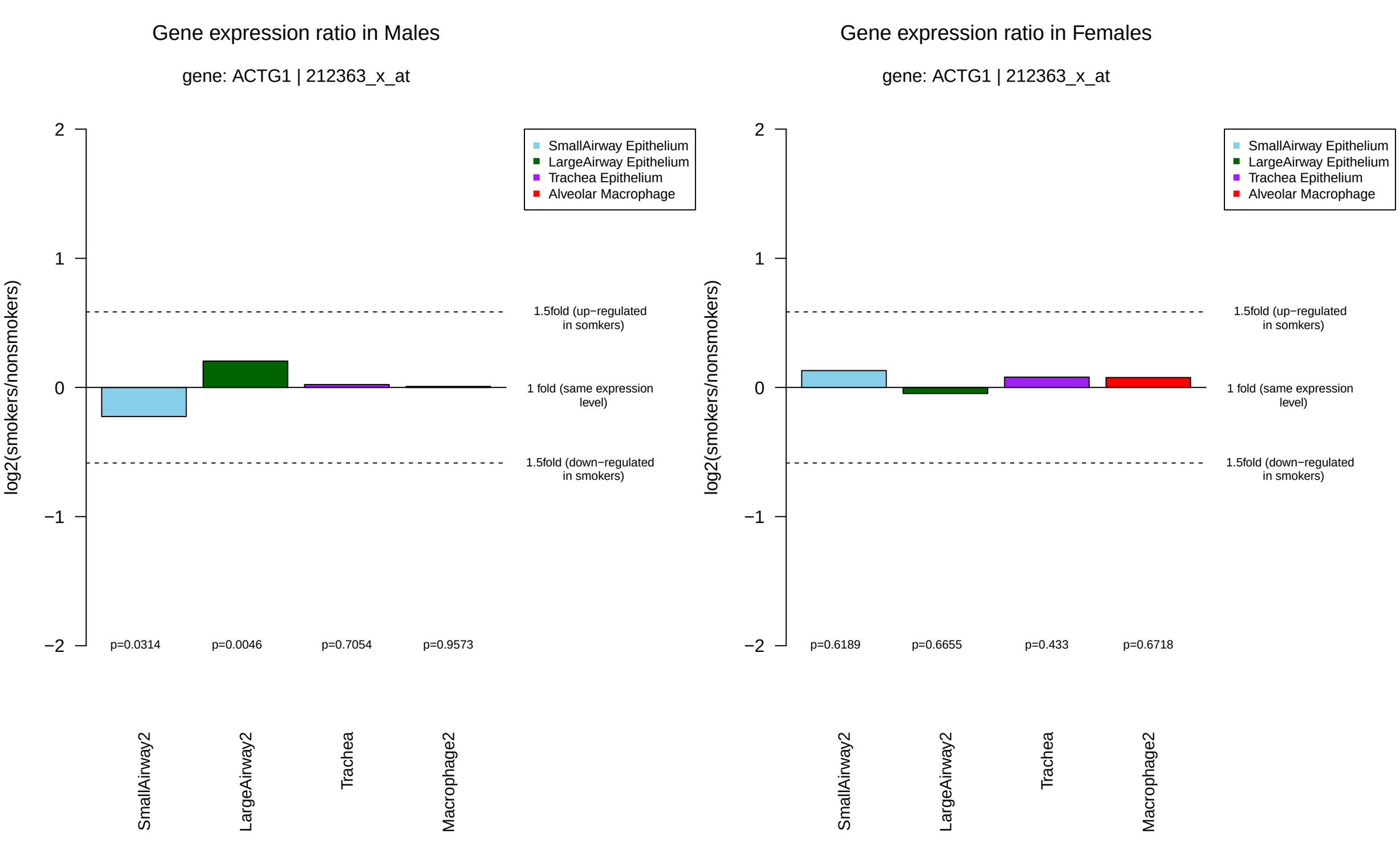
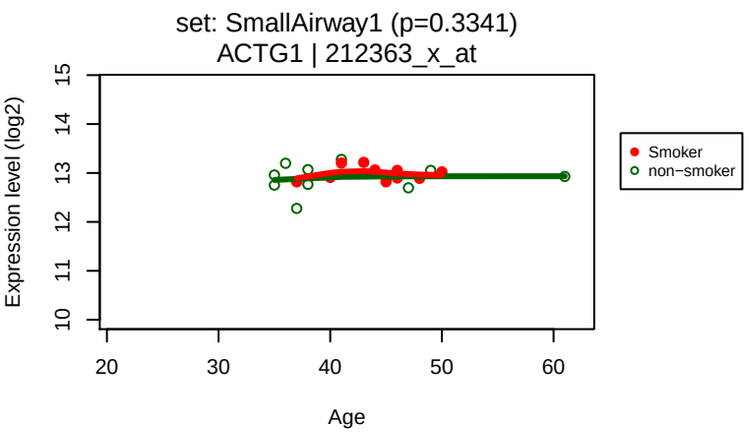
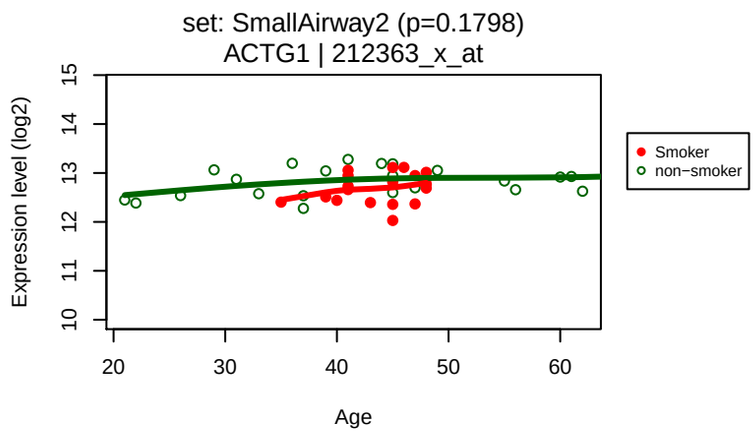
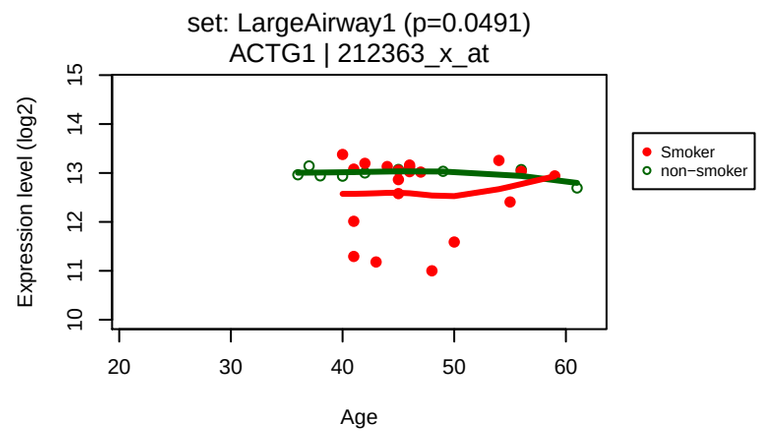

Gene expression of smokers and nonsmokers
gene: ACTG1 | 212363_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3341
p=0.1798
p=0.8418
p=0.0491
p=0.1092
p=0.7558
p=0.4572
p=0.7846
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

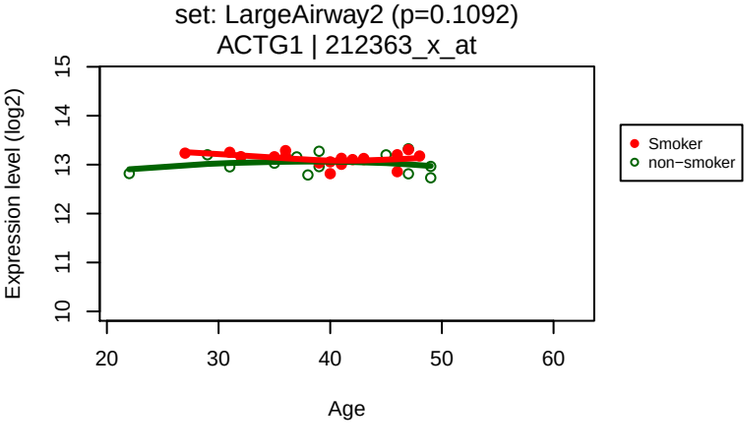

Gene expression ratio (smokers/nonsmokers)
gene: ACTG1 | 212363_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3341
p=0.1798
p=0.8418
p=0.0491
p=0.1092
p=0.7558
p=0.4572
p=0.7846
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

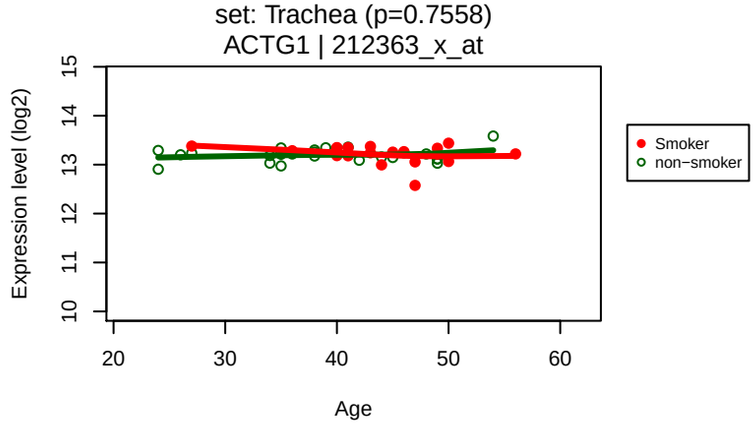

Gene expression ratio in Males
gene: ACTG1 | 212363_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0314
p=0.0046
p=0.7054
p=0.9573
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTG1 | 212363_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6189
p=0.6655
p=0.433
p=0.6718
SmallAirway2
LargeAirway2
Trachea
Macrophage2

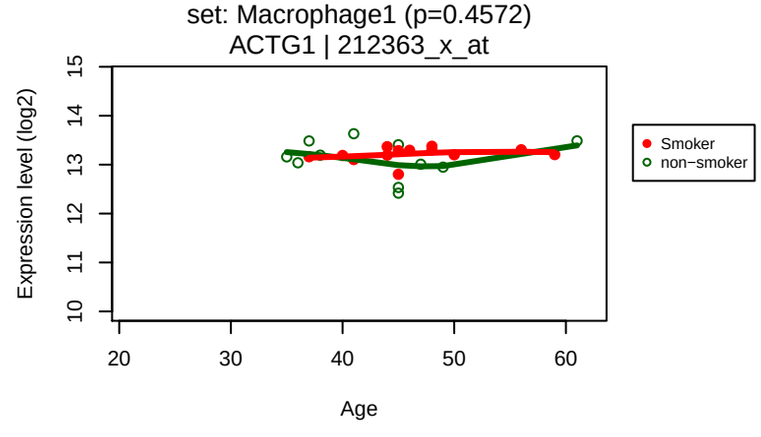

set: SmallAirway1 (p=0.3341)
ACTG1 | 212363_x_at
Expression level (log2)
Age
Smoker
non−smoker

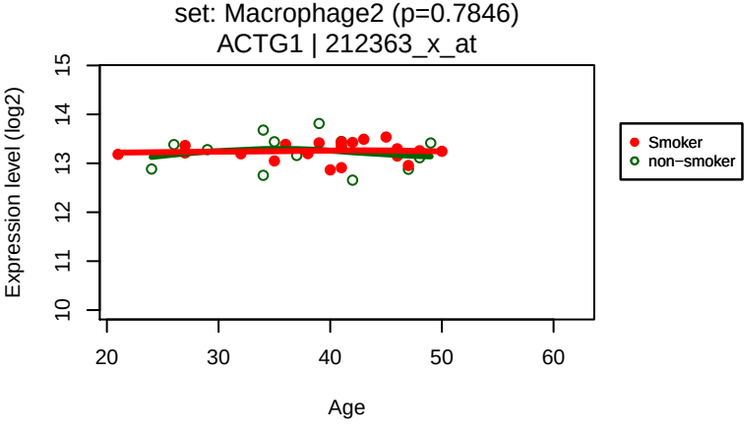

set: SmallAirway2 (p=0.1798)
ACTG1 | 212363_x_at
Expression level (log2)
Age
Smoker
non−smoker

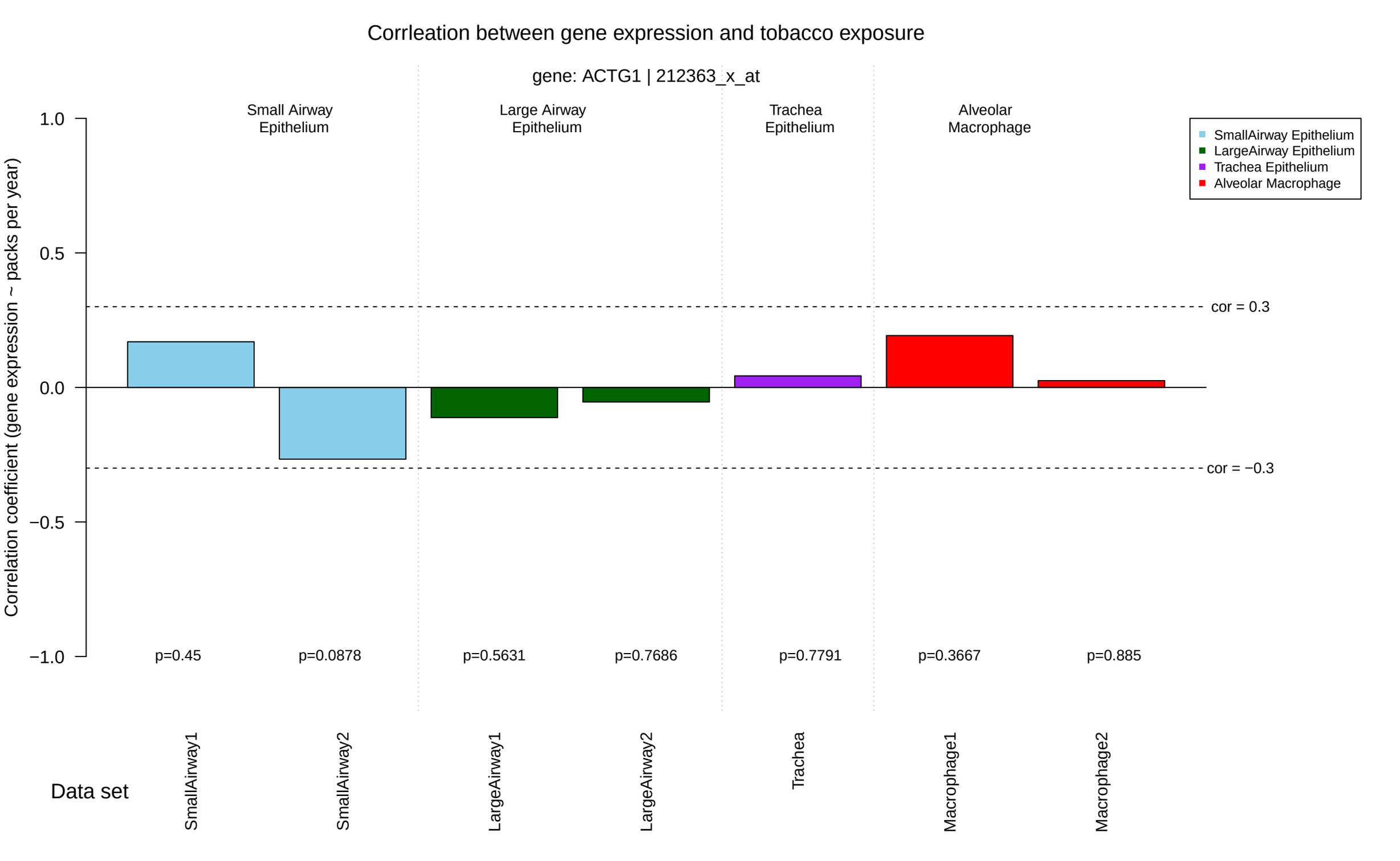

set: LargeAirway1 (p=0.0491)
ACTG1 | 212363_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1092)
ACTG1 | 212363_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7558)
ACTG1 | 212363_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4572)
ACTG1 | 212363_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7846)
ACTG1 | 212363_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTG1 | 212363_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.45
p=0.0878
p=0.5631
p=0.7686
p=0.7791
p=0.3667
p=0.885
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2